

OpenCart API — Test Documentation

Recommended Execution Order

- 1. Module 1 — Authentication → POST api/login (get api_token first)
 - 2. Module 2 — Cart → Add / Get / Edit / Remove products
 - 3. Module 3 — Customer → Set customer details
 - 4. Module 4 — Order → Place & verify orders
 - 5. Module 5 — Token Validation → Confirm auth enforcement
 - 6. Module 6 — Shipping → Set shipping address & method
-

Module 1 — Authentication

Handles API session creation. A successful login returns an api_token that must be passed to every subsequent request. OpenCart always returns HTTP 200; the real status is inside responseCode in the body.

TC-001 · Login — Valid Credentials

Endpoint: POST route=api/login

Priority: High | **Type:** Positive

Preconditions:

- OpenCart API is enabled in Admin → System → Settings → API
- A valid API account exists with status = Enabled
- Environment variables {{api_username}} and {{api_key}} are configured
- Base URL {{base_url}} is reachable

Request Parameters:

username (string, required) — {{api_username}}

key (string, required) — {{api_key}}

Feature: User Authentication via POST api/login

Scenario: Successfully login with valid credentials

Given the OpenCart API is enabled and valid credentials are configured

When I send a POST request to "api/login" with valid username and key

Then the HTTP response status code should be 200

And the response body should contain the key "api_token"

And the api_token value should be a non-empty string

And the api_token is saved to {{api_token}} for downstream tests

Expected Response: { "api_token": "<non-empty-string>" }

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

TC-002 · Login — Invalid API Key

Endpoint: POST route=api/login

Priority: High | **Type:** Negative

Preconditions:

- OpenCart API is enabled and accessible
- The username 'Default' exists but the key 'INVALIDKEY123' does NOT match

Request Parameters:

username (string, required) — Default

key (string, required) — INVALIDKEY123 (intentionally wrong)

Feature: User Authentication via POST api/login

Scenario: Login with an invalid API key returns an error

Given the API is accessible and the username 'Default' exists

When I send a POST request to "api/login" with a valid username but incorrect key

Then the HTTP response status code should be 200

And the response body should contain the key "error"

And the error message should include the word "Incorrect"

And "api_token" key must NOT be present in the response

Expected Response: { "error": { "warning": "Incorrect login or password..." } }

Actual Result: ✓ Status 200 | ✗ Returns empty array [] | AssertionError: expected [] to have property 'error'

Status: **FAIL**

TC-003 · Login — Missing Username Field

Endpoint: POST route=api/login

Priority: High | **Type:** Negative

Request Parameters:

key (string, required) — {{api_key}} (username field omitted entirely)

Feature: User Authentication via POST api/login

Scenario: Login without username field returns an error

- Given** the API is accessible and no prior authentication is needed
- When** I send a POST request to "api/login" omitting the username field entirely
- Then** the HTTP response status code should be 200
- And** the response body should contain the key "error"
- And** "api_token" key must NOT be present in the response

Expected Response: { "error": "..." } — indicates authentication failure

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

TC-003B · Login — Empty Credentials

Endpoint: POST route=api/login

Priority: Medium | **Type:** Negative

Request Parameters:

- username** (string, required) — (empty string)
- key** (string, required) — (empty string)

Feature: User Authentication via POST api/login

Scenario: Login with blank username and key is rejected gracefully

- Given** the API is accessible
- When** I send a POST request to "api/login" with both username and key as empty strings
- Then** the HTTP response status code should be 200
- And** the response body should contain the key "error"
- And** "api_token" key must NOT be present
- And** the system must reject blank credentials gracefully without creating a session

Expected Response: { "error": "..." } — no api_token returned

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

Module 2 — Cart

Full CRUD on the shopping cart. All endpoints require a valid `api_token`. Add a product first (TC-004) to obtain a `cart_id` before running edit or remove tests.

TC-004 · Cart — Add Product Successfully

Endpoint: POST route=`api/cart/add`

Priority: High | **Type:** Positive

Preconditions:

- User is authenticated — `api_token` is valid
- A valid `product_id` exists in the catalogue

Request Parameters:

api_token (string, required) — `{{api_token}}` (query string)

product_id (integer, required) — `{{product_id}}`

quantity (integer, required) — 1

Feature: Cart Management via POST `api/cart/add`

Scenario: Successfully add a product to the cart

Given the user is authenticated and a valid `product_id` exists

When I send a POST request to "`api/cart/add`" with valid `product_id` and quantity

Then the HTTP response status code should be 200

And the response body should contain a "success" property

And the success message should include "modified your shopping cart"

Expected Response: { "success": "Success: You have modified your shopping cart!" }

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

TC-005 · Cart — Add Invalid `product_id`

Endpoint: POST route=`api/cart/add`

Priority: High | **Type:** Negative

Request Parameters:

api_token (string, required) — `{{api_token}}`

product_id (integer, required) — 99999 (non-existent)

quantity (integer, required) — 1

Feature: Cart Management via POST api/cart/add

Scenario: Adding a non-existent product returns an error

Given the user is authenticated

When I send a POST request to "api/cart/add" with a non-existent product_id (99999)

Then the HTTP response status code should be 200

And the response body should contain the key "error"

And the error message should reference "Product"

Expected Response: { "error": { "warning": "Products ... Product" } }

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

TC-006 · Cart — Add Product Without Quantity

Endpoint: POST route=api/cart/add

Priority: Medium | **Type:** Negative

Request Parameters:

api_token (string, required) — {{api_token}}

product_id (integer, required) — {{product_id}}

Note: quantity field is omitted entirely.

Feature: Cart Management via POST api/cart/add

Scenario: Adding a product without quantity field returns a validation error

Given the user is authenticated and a valid product_id exists

When I send a POST request to "api/cart/add" without the quantity field

Then the HTTP response status code should be 200

And the response body should contain the key "error"

And the error message should reference "Quantity"

Expected Response: { "error": { "warning": "... Quantity ..." } }

Actual Result: ✓ Status 200 | ✗ API returns success — product added with no validation |
AssertionError: expected { Object (success) } to have property 'error'

Status: **FAIL**

TC-007 · Cart — Retrieve Cart with Products

Endpoint: GET route=api/cart/products

Priority: High | **Type:** Positive

Preconditions:

- User is authenticated
- At least one product has been added to the cart

Feature: Cart Retrieval via GET api/cart/products

Scenario: Successfully retrieve cart contents

Given the user is authenticated and at least one product is in the cart

When I send a GET request to "api/cart/products"

Then the HTTP response status code should be 200

And the response body should contain a "products" array with at least one item

And the cart_id is saved to {{cart_id}} for downstream tests

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

TC-008 · Cart — Retrieve Empty Cart

Endpoint: GET route=api/cart/products

Priority: Medium | **Type:** Negative

Preconditions:

- User is authenticated and the cart should be empty

Feature: Cart Retrieval via GET api/cart/products

Scenario: Retrieving an empty cart returns an empty products array

Given the user is authenticated and the cart is empty

When I send a GET request to "api/cart/products"

Then the HTTP response status code should be 200

And the response body should contain a "products" array that is empty

Expected Response: { "products": [] }

Actual Result: ✓ Status 200 | ✗ Cart contains leftover items from previous session (cart_id: 17) |
AssertionError: expected array to be empty

Status: **FAIL**

TC-009 · Cart — Edit Product Quantity

Endpoint: POST route=api/cart/edit

Priority: High | **Type:** Positive

Preconditions:

- User is authenticated
- Product is already in the cart with a valid cart_id

Request Parameters:

api_token (string, required) — {{api_token}}

key (string, required) — {{cart_id}}

quantity (integer, required) — 3

Feature: Cart Edit via POST api/cart/edit

Scenario: Successfully update a product quantity in the cart

Given the user is authenticated and a product is in the cart with a valid cart_id

When I send a POST request to "api/cart/edit" with a valid cart_id and quantity 3

Then the HTTP response status code should be 200

And the response body should contain a "success" property

And the success message should include "modified your shopping cart"

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

TC-010 · Cart — Edit Without Key Field

Endpoint: POST route=api/cart/edit

Priority: Medium | **Type:** Negative

Request Parameters:

api_token (string, required) — {{api_token}}

quantity (integer, required) — 2

Note: key field (cart_id) is omitted entirely.

Feature: Cart Edit via POST api/cart/edit

Scenario: Editing cart without key field returns a JSON error response

Given the user is authenticated

When I send a POST request to "api/cart/edit" without the key field

Then the HTTP response status code should be 200

And the response body should contain the key "error" as valid JSON

Expected Response: { "error": "..." } — valid JSON error

Actual Result: ✓ Status 200 | ✗ Server returns raw HTML PHP warning, not JSON | JSONError: Unexpected token '<' | Exposes internal server path

Status: **FAIL**

TC-011 · Cart — Remove Product

Endpoint: POST route=api/cart/remove

Priority: High | **Type:** Positive

Preconditions:

- User is authenticated
- Product is in the cart with a valid cart_id

Request Parameters:

api_token (string, required) — {{api_token}}

key (string, required) — {{cart_id}}

Feature: Cart Removal via POST api/cart/remove

Scenario: Successfully remove a product from the cart

Given the user is authenticated and a product exists in the cart with a valid cart_id

When I send a POST request to "api/cart/remove" with the valid cart_id

Then the HTTP response status code should be 200

And the response body should contain a "success" property

And the success message should include "modified your shopping cart"

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

Module 3 — Customer

Sets customer details on the current API session. Required before placing an order. All endpoints require a valid api_token.

TC-012 · Customer — Add Customer Details

Endpoint: POST route=api/customer

Priority: High | **Type:** Positive

Request Parameters:

api_token (string, required) — {{api_token}}

firstname (string, required) — John

lastname (string, required) — Doe

email (string, required) — john.doe@test.com

telephone (string, required) — 01000000000

Feature: Customer Management via POST api/customer

Scenario: Successfully set customer details on the session

Given the user is authenticated

When I send a POST request to "api/customer" with all required customer fields

Then the HTTP response status code should be 200

And the response body should contain a "success" property

And the success message should include "modified customers"

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

TC-013 · Customer — Add Without Email

Endpoint: POST route=api/customer

Priority: High | **Type:** Negative

Request Parameters:

api_token (string, required) — {{api_token}}

firstname (string, required) — John

lastname (string, required) — Doe

telephone (string, required) — 01000000000

Note: email field is omitted entirely.

Feature: Customer Management via POST api/customer

Scenario: Omitting the email field returns a validation error

Given the user is authenticated

When I send a POST request to "api/customer" without the email field

Then the HTTP response status code should be 200

And the response body should contain an error.warning property

And the warning message should reference "E-Mail"

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

Module 4 — Order

Places orders via the API. Before calling order/add, the session must have: a product in the cart, customer details, a shipping address, a shipping method, a payment address, and a payment method — all set via pre-request scripts.

TC-014 · Order — Place Order with All Details

Endpoint: POST route=api/order/add

Priority: Critical | **Type:** Positive

Preconditions:

- User is authenticated
- Cart has at least one product
- Customer info is set (TC-012)
- Shipping address is set (TC-020)
- Shipping method is selected
- Payment address is set
- Payment method is selected

Feature: Order Placement via POST api/order/add

Scenario: *Successfully place an order when all prerequisites are met*

Given all pre-request conditions are satisfied — cart, customer, shipping, and payment are all configured

When I send a POST request to "api/order/add"

Then the HTTP response status code should be 200

And the response body should contain an "order_id" field

And the order_id is saved to {{order_id}} for downstream use

Expected Response: { "order_id": <number> }

Actual Result: ✓ Status 200 | ✗ Returns error object — order not created | AssertionError: expected { Object (error) } to have property 'order_id'

Status: **FAIL**

TC-015 · Order — Place Order with Empty Cart

Endpoint: POST route=api/order/add

Priority: High | **Type:** Negative

Preconditions:

- User is authenticated
- Cart is empty — no products added

Feature: Order Placement via POST api/order/add

Scenario: *Attempting to place an order with an empty cart returns an error*

Given the user is authenticated and the cart is empty

When I send a POST request to "api/order/add"

Then the HTTP response status code should be 200

And the response body should contain an error.warning property

And the warning message should include "empty"

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

Module 5 — Token Validation

Verifies that all protected endpoints correctly enforce api_token authentication. Missing or invalid tokens must return a permission error.

TC-016 · Token — Request Without api_token

Endpoint: GET route=api/cart/products (no api_token)

Priority: High | **Type:** Negative

Feature: Token Enforcement on GET api/cart/products

Scenario: *Request without api_token is rejected with a permission error*

Given no api_token is included in the request

When I send a GET request to "api/cart/products" without any api_token parameter

Then the HTTP response status code should be 200

And the response body should contain an error.warning property

And the warning message should include "permission"

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

TC-017 · Token — Request with Invalid api_token

Endpoint: GET route=api/cart/products (api_token=abc123invalid)

Priority: High | **Type:** Negative

Feature: Token Enforcement on GET api/cart/products

Scenario: *Request with an invalid api_token is rejected with a permission error*

Given an invalid api_token "abc123invalid" is used

When I send a GET request to "api/cart/products?api_token=abc123invalid"

Then the HTTP response status code should be 200

And the response body should contain an error.warning property

And the warning message should include "permission"

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

TC-018 · Token — Change Currency with Valid Token

Endpoint: POST route=api/currency

Priority: Medium | **Type:** Positive

Request Parameters:

api_token (string, required) — {{api_token}}

currency (string, required) — USD

Feature: Currency Update via POST api/currency

Scenario: *Successfully change the session currency with a valid token*

Given the user is authenticated with a valid api_token

When I send a POST request to "api/currency" with currency=USD

Then the HTTP response status code should be 200

And the response body should contain a "success" property

And the success message should include "currency"

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

TC-019 · Token — Change Currency Without api_token

Endpoint: POST route=api/currency (no api_token)

Priority: High | **Type:** Negative

Request Parameters:

currency (string, required) — USD

Note: api_token is omitted.

Feature: Currency Update via POST api/currency

Scenario: Currency change without api_token is rejected with a permission error

Given no api_token is included in the request

When I send a POST request to "api/currency" without api_token

Then the HTTP response status code should be 200

And the response body should contain the key "error"

And the error message should include "permission"

Actual Result: ✓ Status 200 | ✓ Response matches expected output

Status: **PASS**

Module 6 — Shipping

Sets the shipping address on the current session. Must be called after adding a product to the cart and before placing an order.

TC-020 · Shipping — Set Shipping Address

Endpoint: POST route=api/shipping/address

Priority: High | **Type:** Positive

Preconditions:

- User is authenticated
- At least one product has been added to the cart via pre-request script

Request Parameters:

api_token (string, required) — {{api_token}}
firstname (string, required) — John
lastname (string, required) — Doe
address_1 (string, required) — 123 Test Street
city (string, required) — Cairo
postcode (string, required) — 12345
country_id (integer, required) — 222 (Egypt)
zone_id (integer, required) — 3563

Feature: Shipping Configuration via POST api/shipping/address

Scenario: Successfully set a shipping address on the session

- Given** the user is authenticated and a product is in the cart
- When** I send a POST request to "api/shipping/address" with a complete address payload
- Then** the HTTP response status code should be 200
- And** the response body should contain a "success" property
- And** the success message should include "Shipping address"

Actual Result: ✓ Status 200 | ✓ Response matches expected output
Status: **PASS**

Test Execution Summary

Total Test Cases: 20
Passed: **15**
Failed: **5**
Pass Rate: 75%

Failed Test Cases:

- TC-002 | Module 1 — Authentication | Login with invalid key returns empty array instead of error
- TC-006 | Module 2 — Cart | Add product without quantity returns success instead of error
- TC-008 | Module 2 — Cart | Empty cart retrieval returns leftover items from previous session
- TC-010 | Module 2 — Cart | Edit cart without key returns raw HTML PHP warning instead
- TC-014 | Module 4 — Order | Place order with all details returns error instead of order_id

Modules Coverage:

- Module 1 — Authentication (TC-001 to TC-003B) — 4 test cases
- Module 2 — Cart (TC-004 to TC-011) — 8 test cases
- Module 3 — Customer (TC-012 to TC-013) — 2 test cases
- Module 4 — Order (TC-014 to TC-015) — 2 test cases
- Module 5 — Token Validation (TC-016 to TC-019) — 4 test cases
- Module 6 — Shipping (TC-020) — 1 test case